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SAN CARLOS
SCIENTIA • VIRTUS • DEVOTIO



AgriKita: A Digital Platform for Farmers and Buyers

A Systems Analysis and Design Proposal

Presented to the Faculty of the
Department of Computer, Information Sciences and Mathematics
University of San Carlos

In Partial Fulfillment
of the Requirements for the
SYSTEMS ANALYSIS AND DESIGN

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1.Introduction

a. Purpose of the System Requirements Document

The purpose of this System Requirements Document is to provide a detailed, unified foundation for the development and implementation of the AgriKita app, a dedicated agricultural e-commerce and information platform tailored to the needs of Filipino farmers and ranchers. This document sets out to articulate the vision, objectives, and essential features of the system, ensuring that everyone involved, from technical development teams and designers to business sponsors, agricultural experts, and community partners, shares a clear, consistent understanding of what the app seeks to accomplish.

By outlining both the functional and non-functional requirements, the document acts as a reference point for specifying what the app must do and how it should perform, covering everything from the core marketplace features and transaction flows to crucial aspects like user security, accessibility, and regulatory compliance. It plays a critical role in guiding cross-functional collaboration throughout the project lifecycle, supporting clear communication, minimizing ambiguity, and providing a benchmark against which progress and success can be measured.

Furthermore, this document underpins AgriKita's broader mission to foster digital transformation in Philippine agriculture, empowering farmers and ranchers through improved market access, real-time weather and news updates, fair trade practices, and sustainable growth. It also ensures that the system's design will be robust, scalable, and flexible enough to adapt to evolving user needs and regulatory frameworks, thereby reducing risk and paving the way for a smooth transition from planning to development, and ultimately, to a successful and impactful deployment. Through its comprehensive approach, this document aspires to be both a roadmap and a contract, aligning all stakeholders to a common set of expectations and driving the realization of AgriKita's vision for the future of Filipino agriculture.

b. Intended Audience and Use

This section identifies the primary stakeholders who will refer to this System Requirements Document, including developers, designers, business sponsors, agricultural experts, and partner organizations, and outlines how each will use the document to guide decisions, ensure alignment with project goals, and support the successful development and implementation of the AgriKita app.

This document is for:

- Investors and project sponsors
- App developers, UI/UX designers
- Agricultural experts and educators
- Marketing and outreach personnel
- Government agencies, NGOs, and agricultural partners

c. Scope of the Proposed System (focus on "To-be")

The **"To-be" system**, AgriKita, is a digital platform that connects Filipino farmers directly with local markets and provides tools for crop planning, weather monitoring, digital payment, and learning. The platform will:

- Enable farmers to **access real-time market prices**
- Provide a **crop planning guide** for optimized planting schedules
- Offer **weather alerts and climate-resilient farming advice**
- Facilitate **direct transactions with buyers through digital wallets**
- Supply **educational content and digital farming tools**

d. Definitions, Acronyms, and Abbreviations

Term	Definition
Agrikita	Name of the digital platform
Farmer/Rancher	One of the users of the app
Marketplace	Digital commerce platform within Agrikita
Weather Alert	Notification on weather related recommendations
Buyer	Registered user purchasing agricultural goods
NGO	Non-Governmental Organization
UI/UX	User Interface / User Experience

e. References (if applicable)

- Department of Agriculture of the Philippines
- Agricultural technology industry reports, 2024–2025
- Best practices in agri e-commerce and digital weather alerting

2. System Overview

a. Problem Statement

Filipino farmers face several persistent challenges that limit their earning potential and long-term sustainability. A major issue is the lack of access to real-time market data, which leads to unfair pricing and forces farmers to rely on intermediaries who often take a significant portion of the profits. Without direct connections to buyers, farmers have little bargaining power and remain dependent on middlemen. Additionally, many lack the digital literacy and tools needed to manage their crops, finances, and market opportunities effectively.

Unpredictable weather conditions and limited access to financial services further compound these problems. Sudden climate changes can severely impact yields, yet most farmers receive little to no timely weather information or guidance on climate-resilient practices. At the same time, many are excluded from affordable credit and insurance options, restricting their ability to invest in better tools or recover from losses. Addressing these gaps through technology and inclusive systems is essential to empower farmers and strengthen the agricultural sector.

b. Proposed System Summary

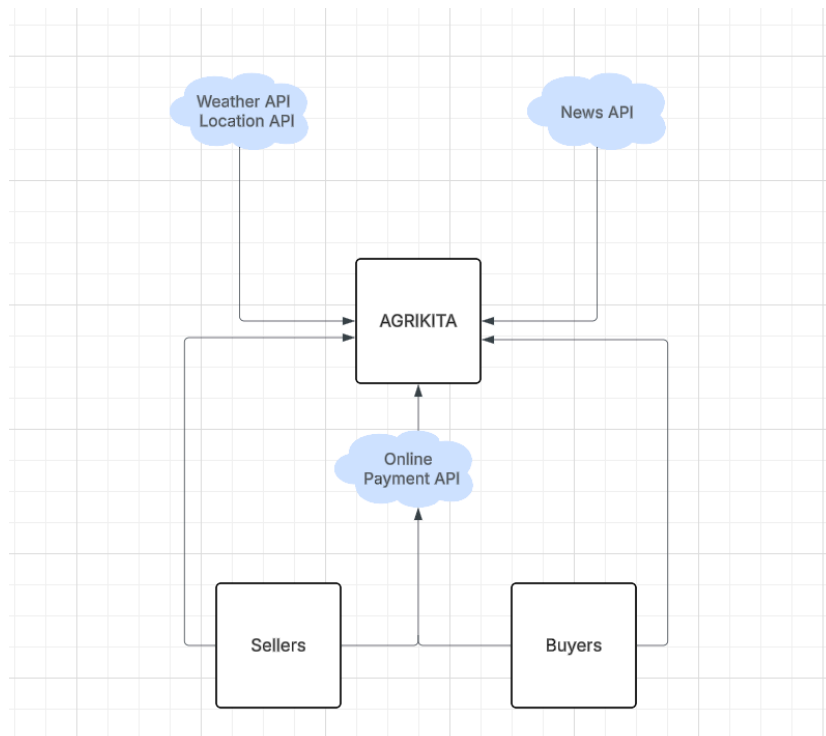
To address the challenges faced by Filipino farmers, AgriKita proposes a mobile and web application designed to provide practical and accessible digital solutions. The platform will feature crop management modules to help farmers plan, monitor, and optimize their planting schedules for better yields. Real-time weather alerts and climate-resilient farming tips will empower farmers to make informed decisions and protect their crops from unpredictable conditions. A market dashboard will display local demand and pricing trends, enabling farmers to align their production with market needs and sell their products at fairer prices.

In addition to data and planning tools, AgriKita will offer a direct communication and ordering system between farmers and buyers, eliminating the need for middlemen and fostering transparent negotiations. The platform will also include curated

educational content and digital tool recommendations to support digital literacy and upskilling. To streamline transactions, AgriKita will integrate mobile wallet services, giving farmers a secure and convenient way to receive payments. Altogether, the system aims to improve farmer income, reduce vulnerability, and build a more connected and resilient agricultural community.

c. System Context Diagram

This part represents the System Context Diagram, which provides a high-level overview of the AgriKita platform and its interactions with external entities. It illustrates how the system connects with users, services, and data sources, clearly defining the system's boundaries and the flow of information between AgriKita and its environment.



d. Assumptions and Constraints

Assumptions

- Farmers have access to smartphones with internet
- Government datasets on pricing are available
- Partner NGOs or agencies assist with onboarding

Constraints

- Unreliable rural internet connectivity
- Varying levels of digital literacy among farmers
- Budget limitations for continuous server/API maintenance

e. Stakeholder Roles and Responsibilities

This section outlines the key stakeholders involved in the AgriKita platform and defines their specific roles and responsibilities in the development, use, and support of the system. It helps clarify who is accountable for various aspects of the project and ensures effective collaboration throughout its lifecycle.

Stakeholder	Role	Responsibility
Users	Clients	Buy and sell agricultural goods.
Agricultural Experts	Content Providers	Provide learning content and techniques
NGOs/Government Agencies	Support	Assist with onboarding and outreach
QA/Testers	Quality Control	Ensure usability and system integrity
AgriKita Dev Team	Developers	Build and maintain the app.

3. Planning Phase

a. Business Needs / Justification

Filipino farmers continue to suffer from low income, poor market access, and a lack of critical digital and weather-related information. These factors severely limit their ability to grow, sustain operations, and improve livelihoods. AgriKita is needed to bridge these gaps by providing a centralized platform where farmers can access market data, connect directly with buyers, manage crops efficiently, and receive guidance and support from agricultural experts, all within a simple, mobile-friendly system.

b. Business Objectives and Success Criteria

Criteria	Target
Farmer Onboarding	At least 1,000 farmers within the first 6 months
User Satisfaction	80% or higher positive feedback in surveys
Increase in Direct Sales	50% of transactions occur directly via the platform
Reduction in Crop Loss Due to Weather	20% decrease reported by farmers using the platform
Platform Uptime and Stability	95% uptime during the first year of operation

Training Completion Rate	90% of registered users complete onboarding/tutorial
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c. Project Scope

i. In-scope Functionalities

- Farmer and buyer user registration/login.
- Crop management and calendar scheduling tools.
- Real-time market dashboard (local prices, demand).
- Direct messaging/order system between buyers and farmers.
- Weather alerts and farming tips module.
- Educational content for digital literacy and farming techniques.
- Integration with mobile wallet/payment systems.
- Admin dashboard for NGO/government partner oversight.

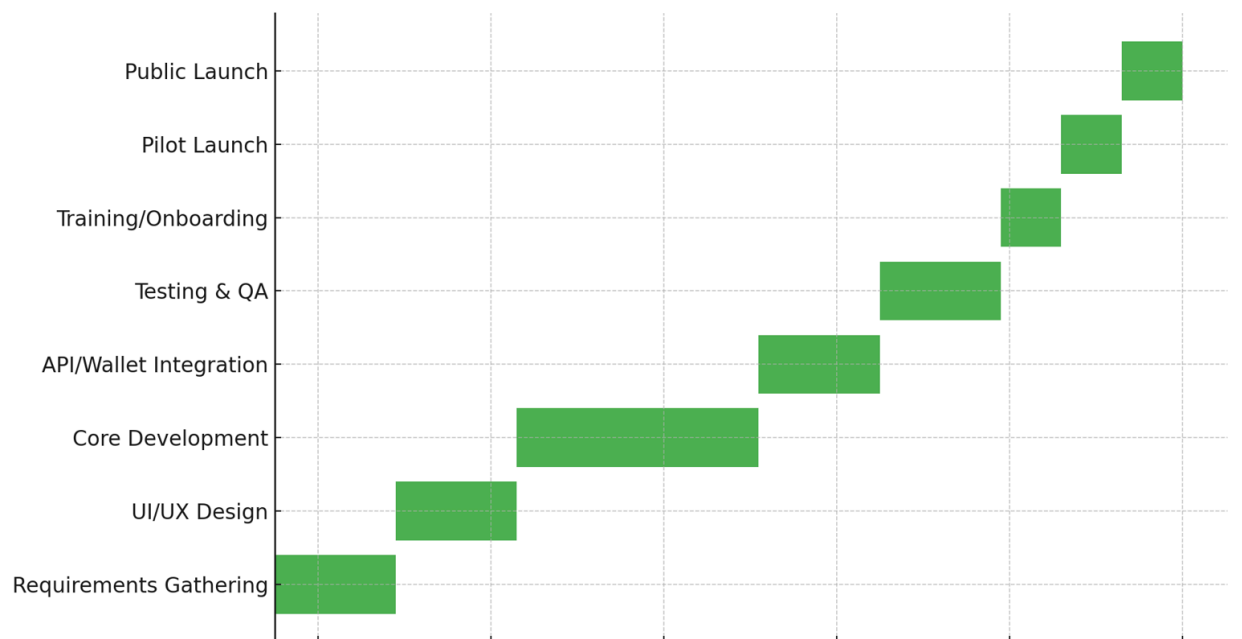
ii. Out-of-scope Items

- Offline access to the full application (initial version requires internet).
- Advanced AI-driven crop prediction tools (future development).
- Physical logistics support (e.g., transport/delivery of goods).
- Full financial service integration (e.g., loans or micro-insurance, unless partnered).
- Multi-language support (limited to Filipino and English initially).

d. Project Risks or Limitations

- Poor internet connectivity in remote areas may hinder adoption.
- Low levels of digital literacy may require extended onboarding/training efforts.
- Incomplete or outdated government market data could affect dashboard accuracy.
- Budget constraints might limit server uptime, scalability, and long-term maintenance.
- Resistance from intermediaries who lose business from direct selling.

e. Estimated Timeline or Milestones



4. Analysis Phase

a. User Roles and Access Levels

Provide a clear mapping of each user type to their permissions, ensuring the right features are accessible to the appropriate stakeholders.

Role	Description	Access Level
Farmer/Rancher	A primary producer who sells goods, views market data, and manages crops.	Create/edit own profile, view marketplace, place listings, access weather alerts, and educational modules.
Buyer	An individual or business purchasing agricultural goods.	Create/Edit profile, browse marketplace, place orders, communicate with sellers.
Guest/Public	An unauthenticated visitor who can browse basic information and news.	View weather forecasts, market-pricing news, prompted to register for full features.
Agricultural Expert	Provides curated content, tips, and recommendations.	Create/edit educational content, view usage analytics.
Delivery Partner	A third-party logistics provider handles order deliveries.	View assigned deliveries; update delivery status, communicate with buyers and farmers.
NGO/Government Agent	Supports onboarding, monitors platform	View farmer lists, usage dashboards, and export reports.

	adoption, and compliance.	
QA/Tester	Quality assurance of platform functionality.	Access to the staging environment, create test cases, and log defects.
Admin	System administrators and developers maintain the app.	Full system rights: manage users, roles, configurations, and monitor logs.

b. User Stories

Capture concise, actionable scenarios that describe how each user interacts with the system to achieve specific goals, driving feature development in sprints.

i. Marketplace Module

1. As a **farmer**, I want to list my produce with quantity, price, and photos so that buyers can easily discover and purchase my products.
2. As a **buyer**, I want to filter listings by location and crop type so that I can find relevant products quickly.
3. As an **Admin**, I want to approve or reject new listings so that only quality products appear on the marketplace.

ii. Crop Planning & Monitoring

1. As a **farmer**, I want to access a crop calendar and planting schedule so that I can optimize my yield throughout the season.

2. As an **agricultural expert**, I want to publish recommended planting guides so that farmers can follow best practices.

iii. **Weather & Alerts**

1. As a **farmer**, I want to receive real-time weather alerts so that I can protect my crops during extreme conditions.
2. As an **admin**, I want to configure alert thresholds and data sources so that weather notifications remain accurate.

iv. **Digital Wallet & Transactions**

1. As a **farmer**, I want to link my mobile wallet so that I can receive payments securely.
2. As a **buyer**, I want to view my transaction history so that I can track my expenses.

v. **Educational Content**

1. As a **farmer**, I want to browse tutorials and articles so that I can improve my farming techniques.
2. As an **agricultural expert**, I want to manage my authored content so that it stays up to date.

c. **Functional Requirements Overview**

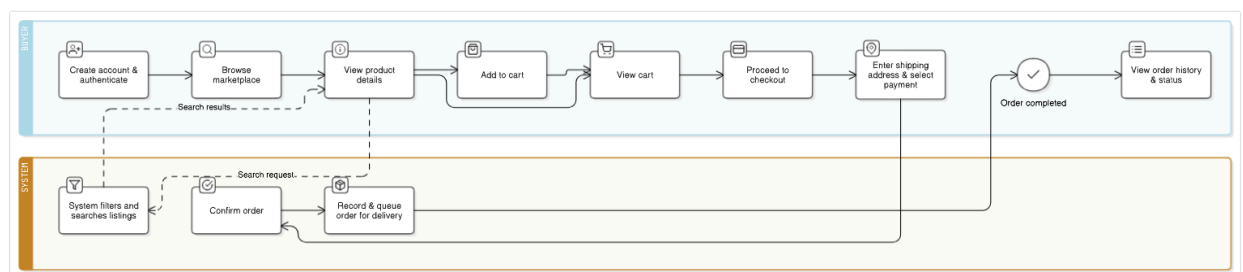
Summarize the core capabilities the system must deliver, covering all modules and ensuring comprehensive feature coverage and performance expectations.

- i. **User Management:** Registration via email or mobile number with OTP verification, secure authentication, and role-based authorization.
- ii. **Marketplace:** CRUD operations for product listings; search, filter, and sort by location, crop type, and price; and in-app messaging between buyer and seller.

- iii. **Weather & News Module:** Interactive weather forecasts (hourly and five-day), news section displaying crop pricing and dry/wet-season temperature data, and real-time weather alerts via push notifications.
- iv. **Payments:** Integration with GCash and PayMaya APIs for secure transaction processing, transaction history, and Cash-on-Delivery (COD)
- v. **Content Management:** Rich-text authoring tool for agricultural experts to create and edit tutorials, content categorization by topic and tags, and user rating and feedback on articles.
- vi. **Reporting & Analytics:** Dashboards for farmers and buyers to review transaction histories and sales trends.
- vii. **Shipping:** Support for shipping options, including assignment of delivery partners and real-time delivery status updates.

d. Use Case Descriptions and Diagrams

The diagram breaks the buyer purchase process into two swimlanes. Buyer (top) and System (bottom) and show how they hand off control at key points:



i. Buyer Lane

1. **Create Account & Authenticate:** The Buyer registers or logs in, proving identity (email/OTP).

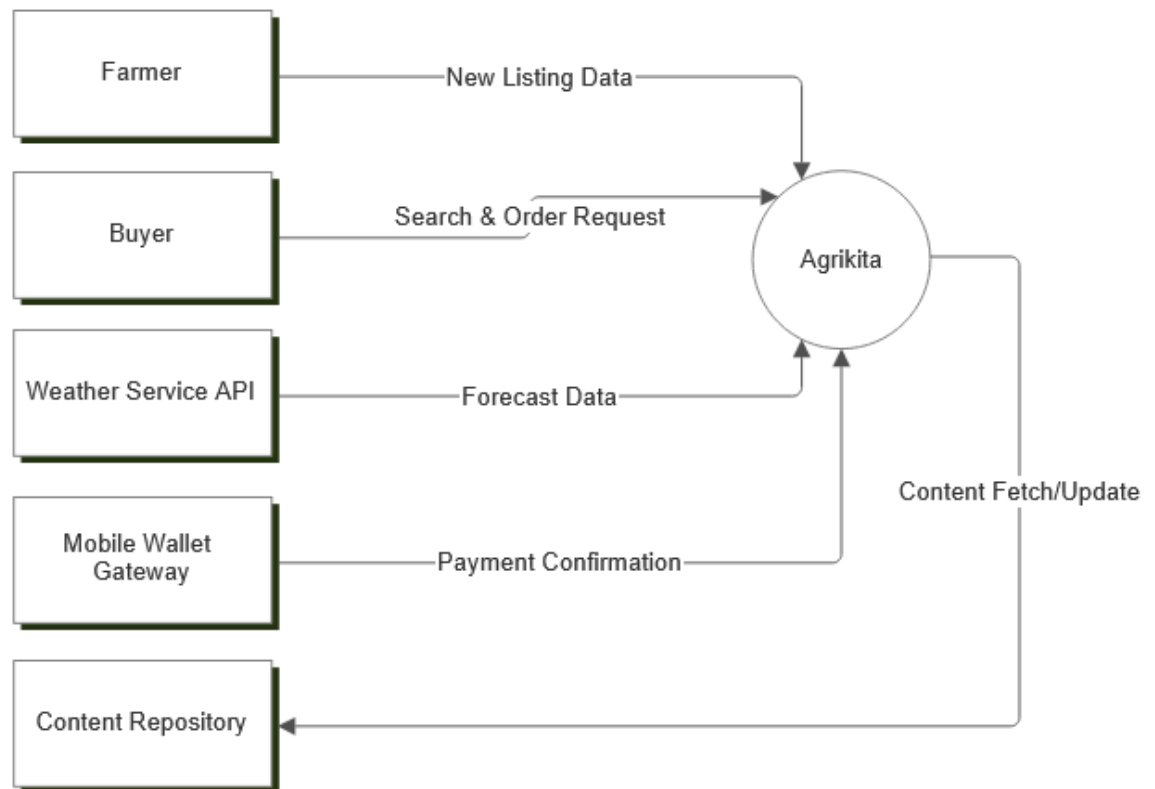
2. Browse Marketplace → View Product Details → View Cart: The Buyer explores listings, refines results, inspects an item, and adds it to the shopping cart.
3. Proceed to Checkout → Enter Shipping Address & Select Payment: The Buyer provides a delivery address and chooses between GCash/PayMaya or Cash-on-Delivery.
4. View Order History & Status: After checkout, the Buyer can check past orders and track their current status.

ii. **System Lane**

1. Filter Search: The System applies the buyer's filter and search criteria, organizing matching products for display.
2. Confirm Order: Once the Buyer submits payment or COD selection, the System validates the transaction and confirms the order.
3. Record & Queue Order for Delivery: The System persists the order in the database, marks it as "queued," and notifies fulfillment or delivery services.

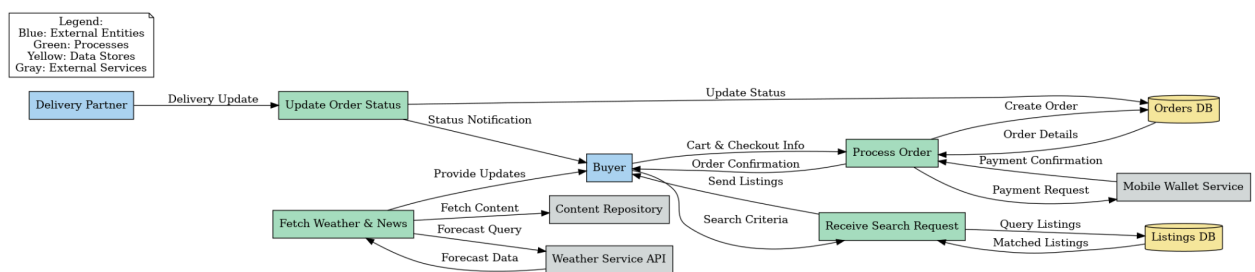
e. **Data Flow Diagrams**

The Data Flow Diagrams are structured into two levels to capture both the system's overall boundary and its detailed internal processes. Level 0 (Context Diagram) provides a bird's-eye view of AgriKita's system boundary and its interactions with external entities. Level 1 (Marketplace Sub-Process) decomposes the core marketplace into detailed subprocesses to show internal data flows. Together, they offer both macro and micro perspectives on how data moves through the platform.



Level 0 (Context Diagram)

This Level 0 context diagram centers on the AgriKita Platform, with Farmers submitting new listing data and Buyers sending search and order requests. It shows the platform querying the Weather Service API for forecast data and receiving payment confirmations from the Mobile Wallet Gateway. Finally, AgriKita fetches and updates content in the Content Repository, illustrating its key high-level interactions and external dependencies.



Level 1 (Marketplace Sub-Process)

This diagram details AgriKita's marketplace engine by mapping how a buyer's search criteria enter 'Receive Search Request,' query the Listings database, and return matched results. It also shows how cart details flow into 'Process Order,' trigger a mobile wallet payment, update the Orders database, and feed back an order confirmation to the buyer

f. Process Descriptions or Workflows

Outline the end-to-end steps and decision points for key user and system interactions such as onboarding, purchasing, and fulfillment using clear, sequential narratives.

i. New User Onboarding

1. Access Platform: User downloads the mobile app or visits the AgriKita web portal.
2. Register & Verify: User completes a registration form with name, email, or phone number; receives and enters OTP to verify identity.
3. Choose Role: User selects their role—Farmer, Buyer, or Agricultural Expert—from a guided menu.
4. Configure Profile: System applies role-based default permissions, prompts user to fill out any additional profile details (e.g., farm location or business info).
5. Orientation Tutorial: System launches a brief, interactive walkthrough of key features relevant to the user's role (marketplace navigation for Buyers, listing tools for Farmers, content tools for Experts).

ii. **Transaction Workflow**

1. Product Selection: Buyer browses listings and adds one or more products to their cart.
2. Checkout Initiation: Buyer reviews cart contents and taps “Checkout.”
3. Payment Processing: Buyer chooses a payment method (linked GCash/PayMaya account or Cash-on-Delivery), enters any required details, and confirms payment.
4. Order Confirmation: Payment gateway processes the transaction; system validates the response and issues an order confirmation to both Buyer and Farmer.
5. Fulfillment & Delivery: Farmer receives the order notification, prepares and dispatches the goods; Farmer updates the order status as “Out for Delivery” or “Ready for Pickup.”
6. Completion & Feedback: Once the Buyer confirms receipt, the system marks the order “Completed,” and both Buyer and Farmer are prompted to rate each other and leave feedback.

5.Design Phase

5.1 System Architecture Overview (high-level component view)

High-Level Component View:

- **Frontend:**
 - React/Flutter-based web & mobile interfaces
 - User login, sign-up, module browsing, and marketplace
- **Backend:**
 - RESTful API (Node.js/Express or Django)
 - Authentication (JWT)
 - Business logic layer
- **Database:**
 - Relational DB (MySQL/PostgreSQL)
 - Stores user data, module content, marketplace listings, and transactions
- **Storage:**
 - Cloud storage for media (e.g., Firebase, AWS S3)
- **Other Components:**
 - Email/SMS service (for password reset, verification)
 - Admin dashboard

5. 2 Module Design (e.g., screens, inputs, outputs per module)

Login/Registration Module

- **Inputs:** Contact Number, Email, Password

- **Outputs:** Login status, validation messages
- **Screens:** Login, Register, Forgot Password

Module Learning

- **Inputs:** Module selection, user interaction
- **Outputs:** Content display, progress (if tracked)
- **Screens:** Module list, Module detail

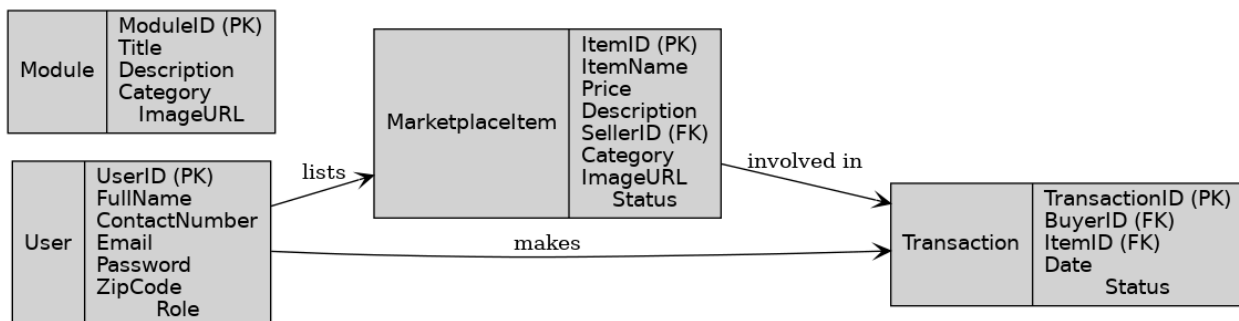
Marketplace

- **Inputs:** Item name, description, price, image
- **Outputs:** Listed items, transaction confirmation
- **Screens:** Item listing, Item detail, Purchase screen

Admin Panel (optional)

1. **Inputs:** Content and user management tools
2. **Outputs:** Metrics, moderation status

5.3 Entity-Relationship Diagram (ERD)



5.4 Data Dictionary / Data Entities

User

Attribute	Type	Description
UserID	INT	Unique identifier for each user (Primary Key)
FullName	VARCHAR(255)	Full name of the user
ContactNumber	VARCHAR(20)	Phone number of the user
Email	VARCHAR(255)	Email address of the user
Password	VARCHAR(255)	Encrypted password
ZipCode	VARCHAR(10)	Postal code representing the location
Role	VARCHAR(50)	User role(Admin, Buyer, Seller)

Module

Attribute	Type	Description
ModuleID	INT	Unique identifier for each module (Primary Key)
Title	VARCHAR(255)	Title of the module
Description	TEXT	Detailed content of the module
Category	VARCHAR(100)	Classification of the module

MarketplaceItem

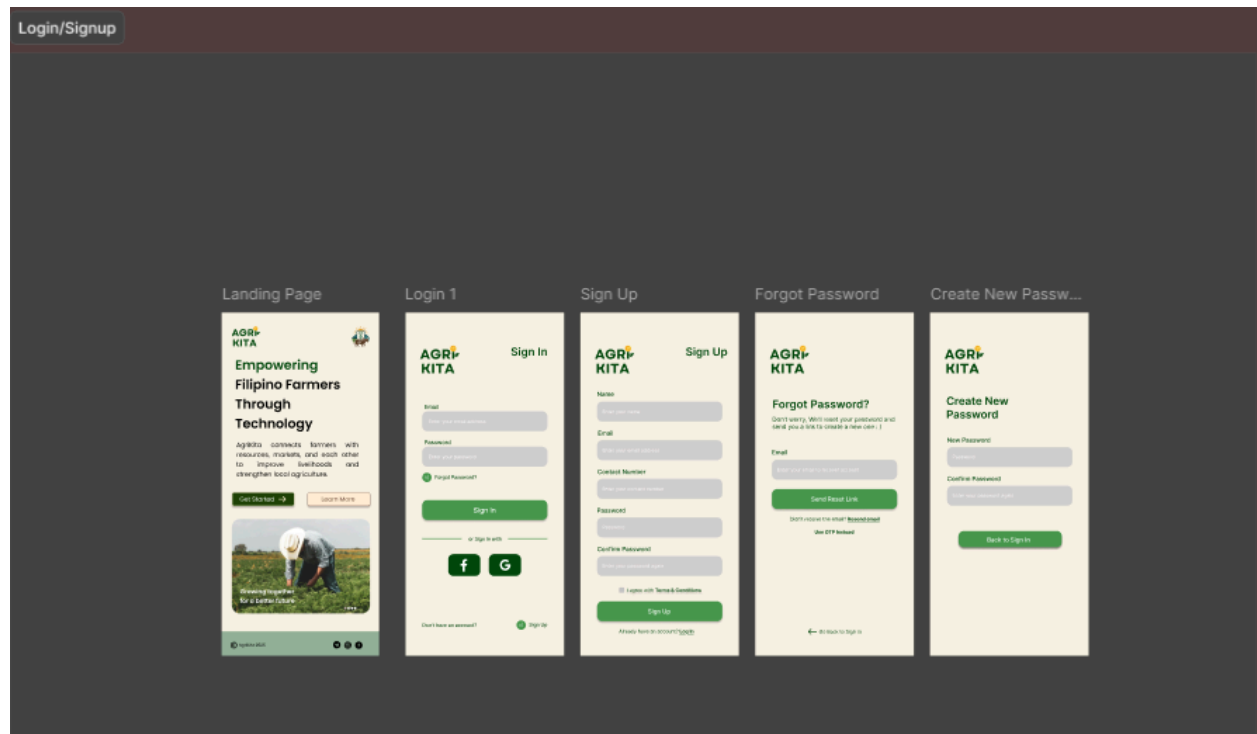
Attribute	Type	Description
ItemID	INT	Unique identifier for each module (Primary Key)
ItemName	VARCHAR(255)	Name of the item being sold
Price	DECIMAL(10,2)	Price of the item
Description	TEXT	Description of the item
SellerID	INT	ID of the user selling the item (Foreign Key)
Category	VARCHAR(100)	Item category/type
Status	VARCHAR(50)	Availability Status(Available, Sold, etc.)

Transaction

Attribute	Type	Description
TransactionID	INT	Unique identifier for each transaction (Primary Key)
BuyerID	INT	ID of the user making the purchase (Foreign Key)
ItemID	INT	ID of the item purchased (Foreign Key)

Date	DATETIME	Date and time of the transaction
Status	VARCHAR(50)	Status of the transaction (Pending, Completed)

User Interface Wireframes/prototype



Weather Forecast

Farming Advice

Good day for field work. Consider irrigation as temperatures are high. Soil conditions are suitable for tillage operations

Humidity: 65% Rainfall: 0mm
Wind: 10 km/h Temp Range: 22°C-28°C

Today's Weather

Talamban, Cebu

28°C

Mostly Sunny



Humidity
65%

Wind
10 km/h

Rainfall
0mm

Farmer's Tip

Good day for field work. Consider irrigation as temperatures are high. Soil conditions are suitable for tillage operations

5-Day Forecast



Today April 30, 2025
Mostly Sunny

28°C



Tomorrow May 1, 2025
Partly Cloudy

26°C



Wednesday May 2, 2025
Rain Showers

25°C



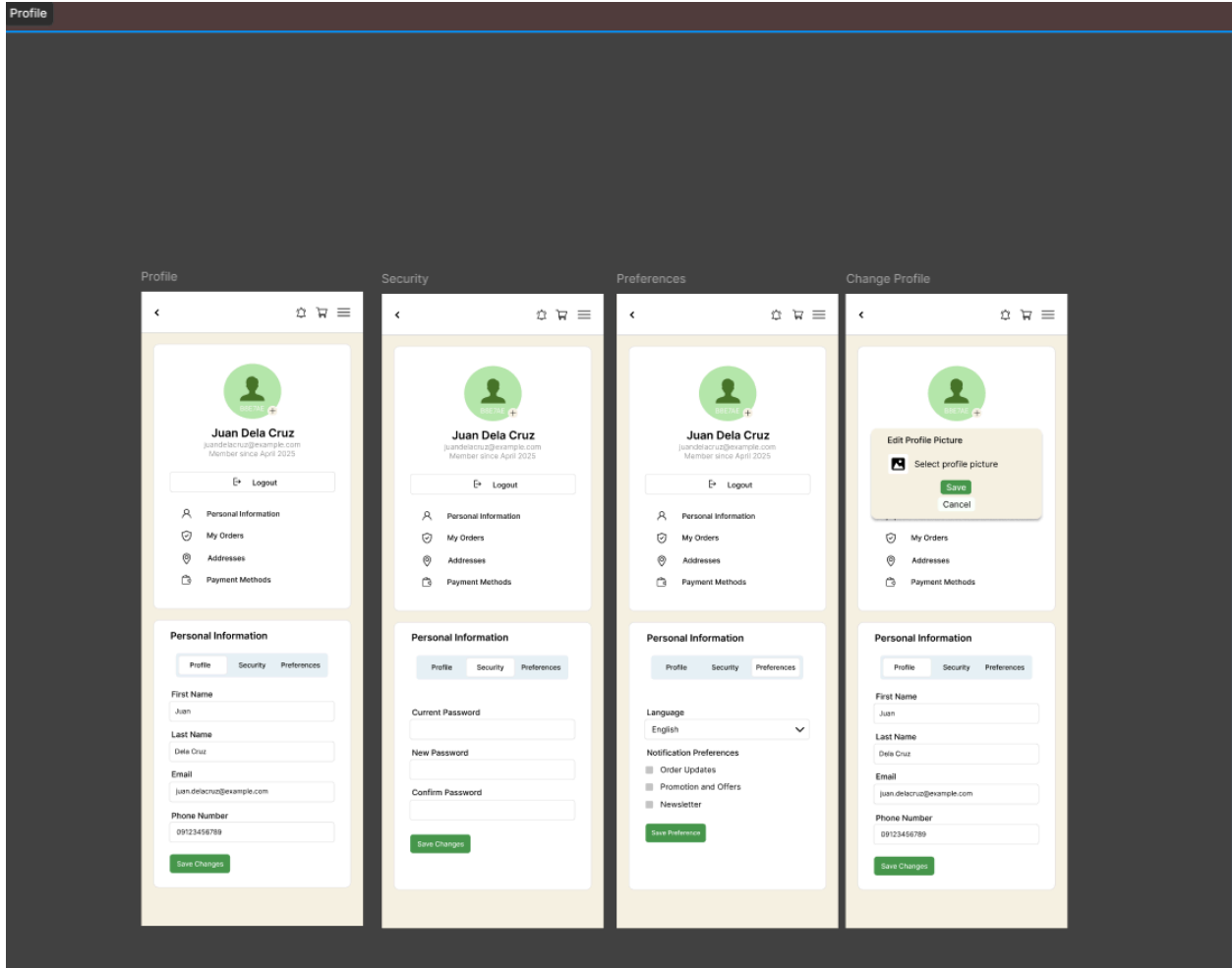
Thursday May 2, 2025
Sunny

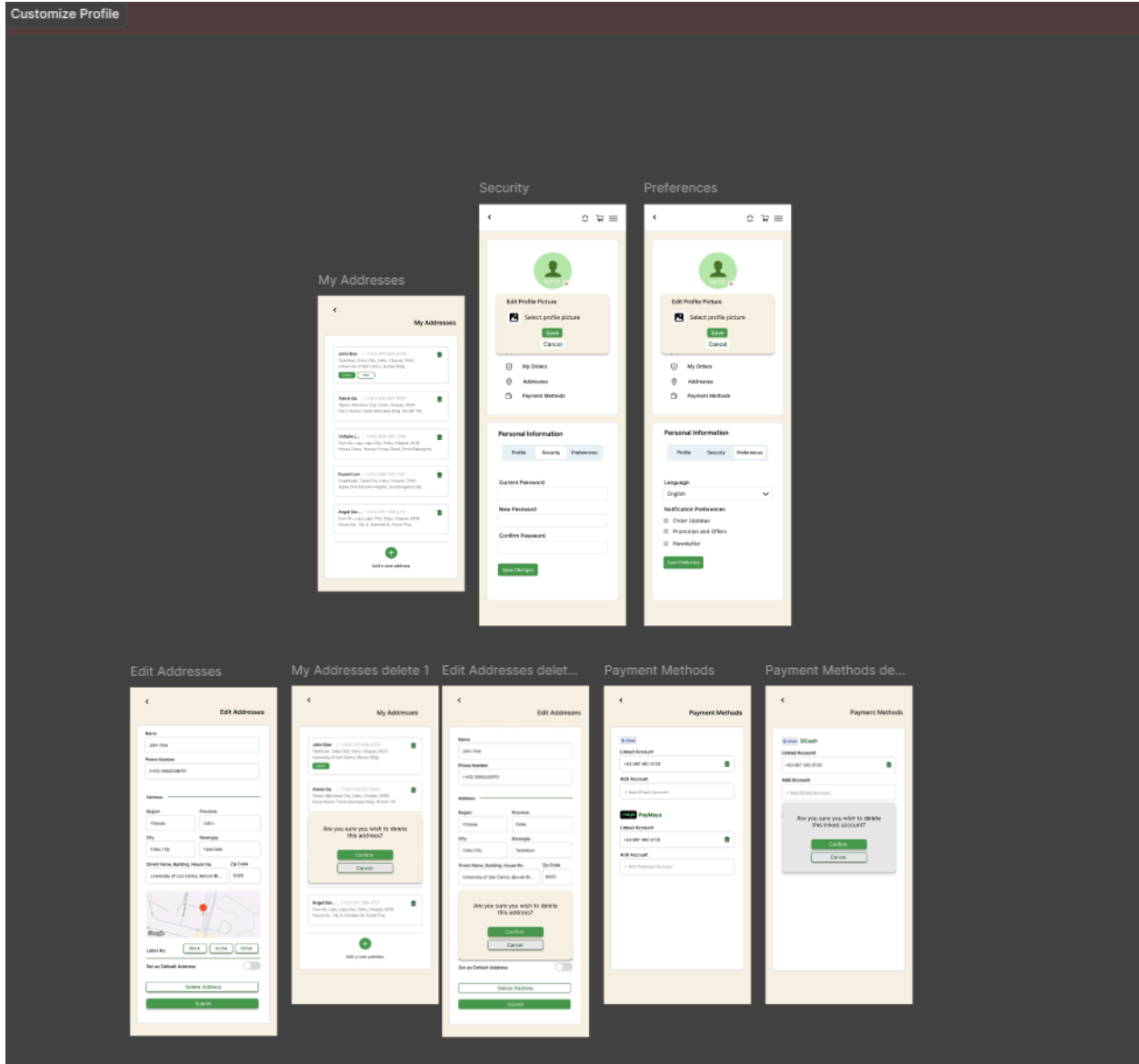
29°C

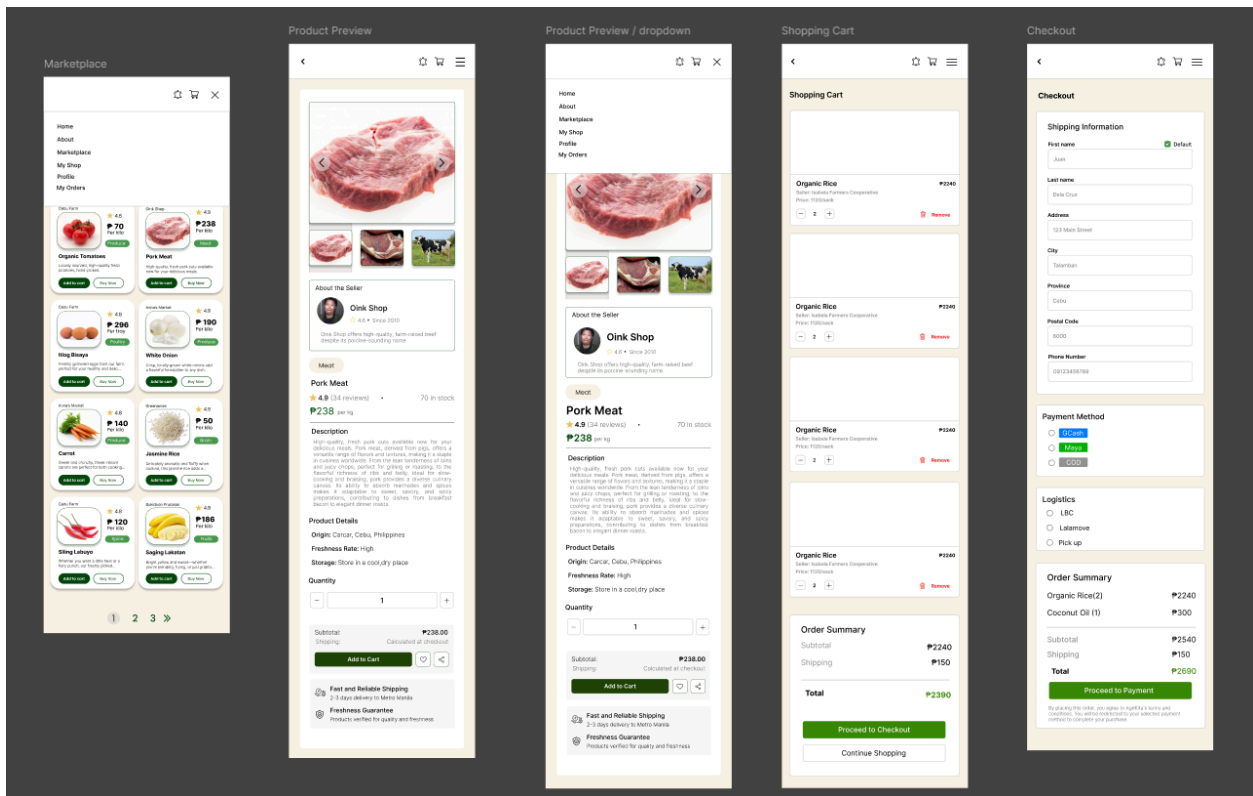
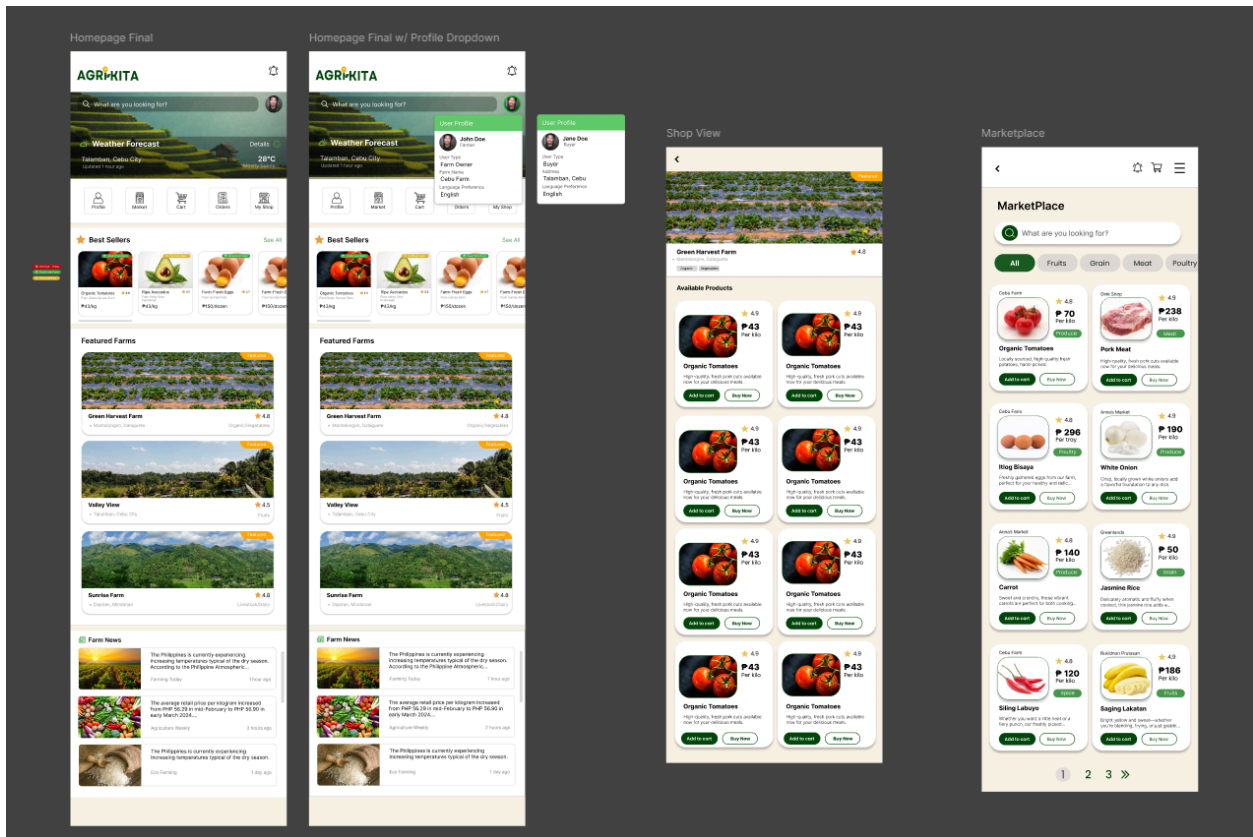


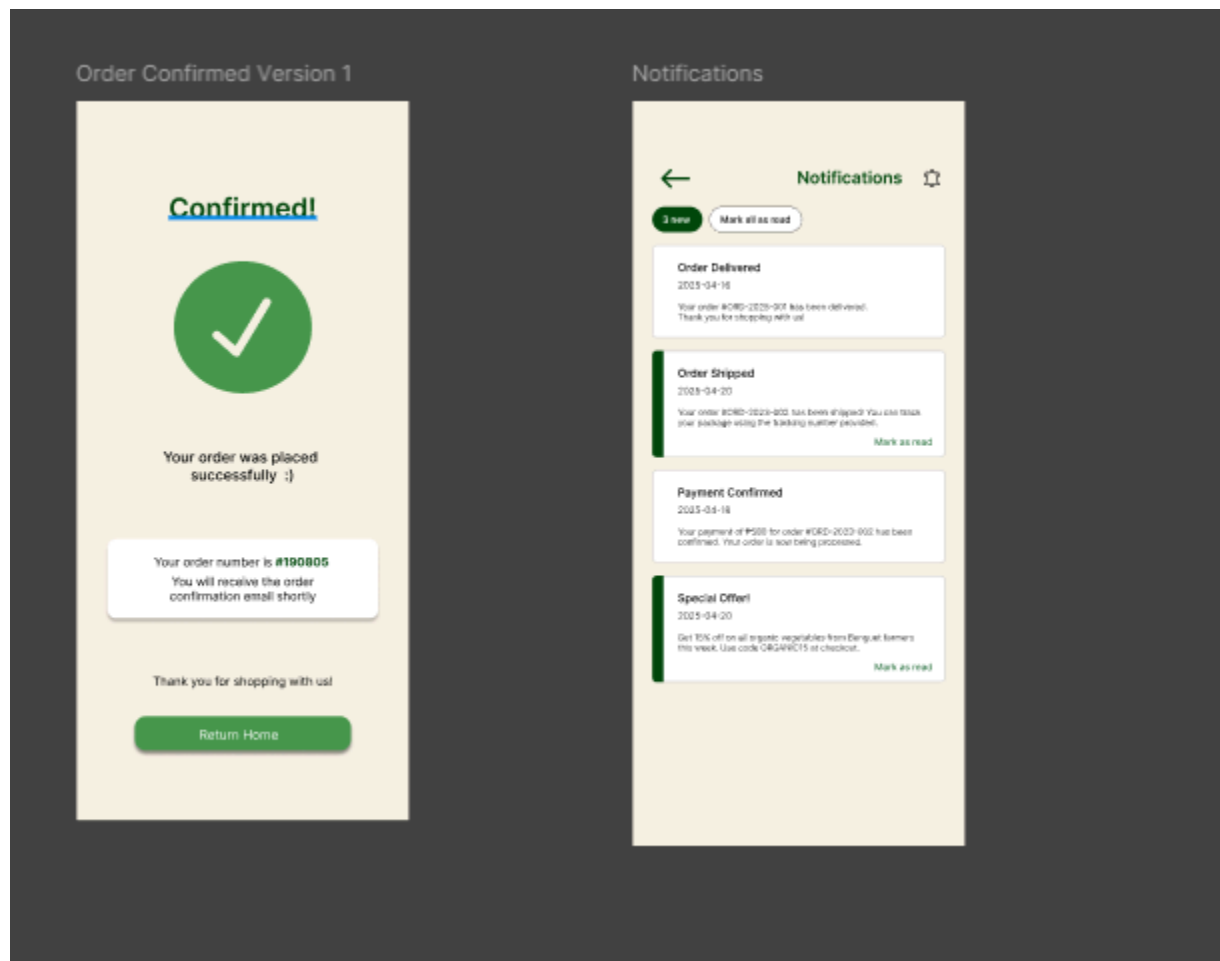
Friday May 2, 2025
Clear Sky

31°C









Navigation Flow or Sitemap

None

Home

└─ Login

└─ Sign Up

└─ Forgot Password

└─ Modules

| └─ Module Detail

└─ Marketplace

| └─ Item Detail

| └─ Purchase Page

└─ Profile

└─ Admin (optional)

6. Appendices

a. Glossary of Terms

Term	Definition
AgriKita	Name of the digital platform for an agriculture-focused marketplace and content.
Farmer	A user who lists, manages, and sells crops through the AgriKita platform.
Buyer	A registered user who purchases goods listed by farmers.
Crop Calendar	A scheduling module for optimal planting and harvesting times.
Weather Alerts	Real-time notifications for weather risks affecting farming.
Digital Wallet	Integration with mobile payment apps (e.g., GCash, PayMaya) for transactions.
Admin	Platform manager with control over users, listings, and site content.
Marketplace Item	Any product (e.g., crop or tool) listed for sale on the platform.
Module	A content unit created by agricultural experts (articles, videos, guides).
Transaction	A record of a purchase including item, buyer, time, and payment.

Delivery Partner	A logistics provider who handles product shipping and updates order status.
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- b. Revision History
- c. Approval/Submission Page (optional or mock-up for academic use)

SRS DOCUMENT FORMAT GUIDELINES

- Paper and Font. Use 8.5" x 11". Use one side of the page only. Use Arial for font style and font size 12-point.
- Margins. Left 1.5 inches; top, bottom, and right, 1 inch.
- Spacing. Use 1.5 spacing for the text. Use single space for table and figure captions.
- Page Number. Show page number at the bottom right side of the page
- Title Page. No page number but counted as page i
- Table of Contents. Counted as page ii. Page number shown